

CHE 201 Spring 2020 HW 6

Roxanna Mendoza

TOTAL POINTS

96.5 / 100

QUESTION 1

1 Question 1 14 / 15

- 0 pts Correct
- ✓ - 1 pts Wrong unit
- 9 pts Wrong volume of air per kilogram
- 6 pts Wrong understanding of volumetric flow rate

factors

- 5 pts Not enough evaluation on economic factors
- 2.5 pts Not enough evaluation on the effects on the ecosystem
- 5 pts Not enough evaluation on the effects on the ecosystem
- 15 pts Not answered

QUESTION 2

2 Question 2 15 / 15

- ✓ - 0 pts Correct

QUESTION 3

3 Question 3 15 / 15

- ✓ - 0 pts Correct
- 2.5 pts a) Wrong calculation
- 2.5 pts a) Wrong equation
- 2.5 pts b) Wrong calculation
- 2.5 pts b) Wrong equation
- 2.5 pts c) Wrong calculation
- 2.5 pts c) Wrong equation

QUESTION 4

4 Question 4 20 / 20

- ✓ - 0 pts Correct

QUESTION 5

5 Question 5 20 / 20

- ✓ - 0 pts Correct

QUESTION 6

6 Question 6 12.5 / 15

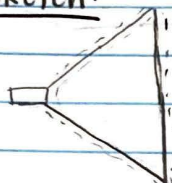
- 0 pts Correct
- ✓ - 2.5 pts Not enough evaluation on technical factors
- 5 pts Not enough evaluation on technical factors
- 2.5 pts Not enough evaluation on economic

Homework #6

Roxanna Mendoza

- #1. Air exists at a turbine at 200 kPa and 150°C with a mass flow rate of 11.5 kg/s. Modeling air as an ideal gas, determine the volumetric flow rate, in kg/s.

Sketch:



$$\dot{m} = 11.5 \text{ kg/s}$$

$$P_1 = 200 \text{ kPa}$$

$$T_1 = 150^\circ\text{C}$$

$$\vec{V} \cdot A = \text{volumetric flow rate}$$

Engineering Model:

① open system/control volume

② ideal gas

$$\dot{m} = \frac{\vec{V} \cdot A}{v} \leftarrow \text{volumetric flow rate}$$

$$P_1 v_1 = RT_1$$

$$\frac{RT_1}{P_1} = v_1$$

$$R = 0.314 \text{ J/mol}\cdot\text{K} \Rightarrow R = 0.314 \frac{\text{N}\cdot\text{m}}{\text{K}}$$

$$T = 150 + 273.15 = 423.15 \text{ K}$$

$$v_{\text{specific}} = \frac{v_1}{M}$$

$$M = 28.97 \text{ kg of air}$$

$$v = \frac{(0.314 \frac{\text{N}\cdot\text{m}}{\text{K}})(423.15 \text{ K})}{(2 \times 10^5 \text{ Pa})(28.97 \text{ kg})} = 0.407 \frac{\text{m}^3}{\text{kg}}$$

$$\dot{m} = \frac{\vec{V} \cdot A}{v}$$

$$11.5 \text{ kg/s} = \frac{\vec{V} \cdot A}{0.407 \text{ m}^3/\text{kg}}$$

$$\vec{V} \cdot A = 6.901 \frac{\text{m}^3}{\text{s}} \Rightarrow 6901 \frac{\text{kg}}{\text{s}}$$

← Volumetric flow rate

1 Question 1 14 / 15

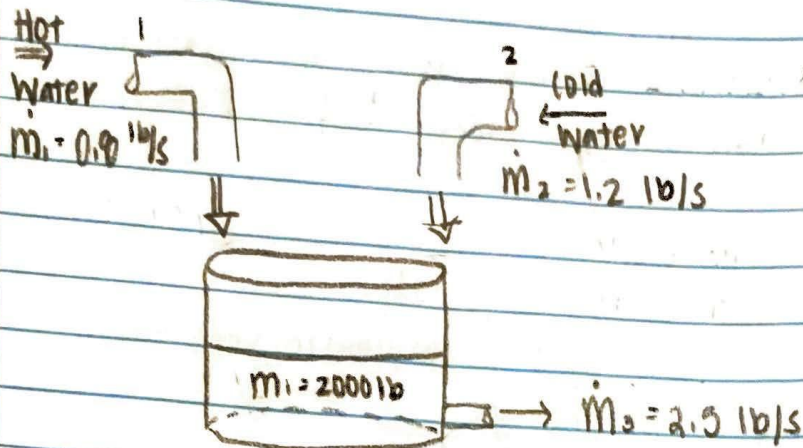
- 0 pts Correct

✓ - 1 pts Wrong unit

- 9 pts Wrong volume of air per kilogram

- 6 pts Wrong understanding of volumetric flow rate

#2.



$m_f = ??$ after one hour

$$\dot{m}_1 = (0.9 \text{ lb/s}) (60 \text{ s}) (60 \text{ min}) = 2900 \text{ lb/hr}$$

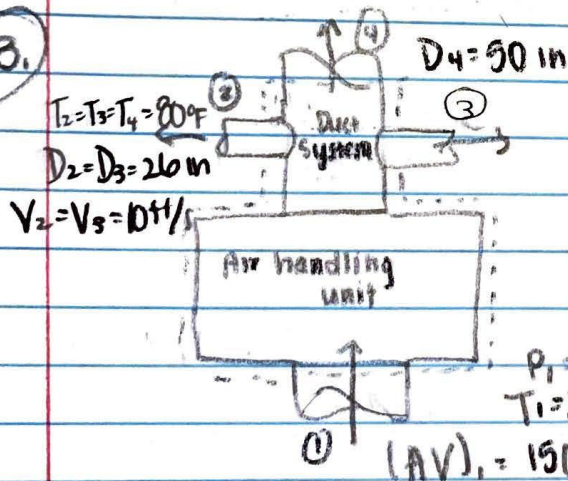
$$\dot{m}_2 = (1.2 \text{ lb/s}) (60 \text{ s}) (60 \text{ min}) = 4320 \text{ lb/hr}$$

$$\dot{m}_3 = (2.5 \text{ lb/s}) (60 \text{ s}) (60 \text{ min}) = 9000 \text{ lb/hr}$$

$$m_f = (2000 \text{ lb}) + (2900 \text{ lb/hr}) + (4320 \text{ lb/hr}) - (9000 \text{ lb/hr})$$

$$m_f = 200 \text{ lb} \leftarrow \text{amount of water after 1 hr}$$

#3.



Engineering model

① open system/control volume

② ideal gas

Part A: $\dot{m}_1 = \frac{AV}{v}$; $T = 35^\circ\text{F} \Rightarrow 274.92 \text{ K}$
 $P = 1 \text{ atm} \Rightarrow 101325 \text{ Pa}$

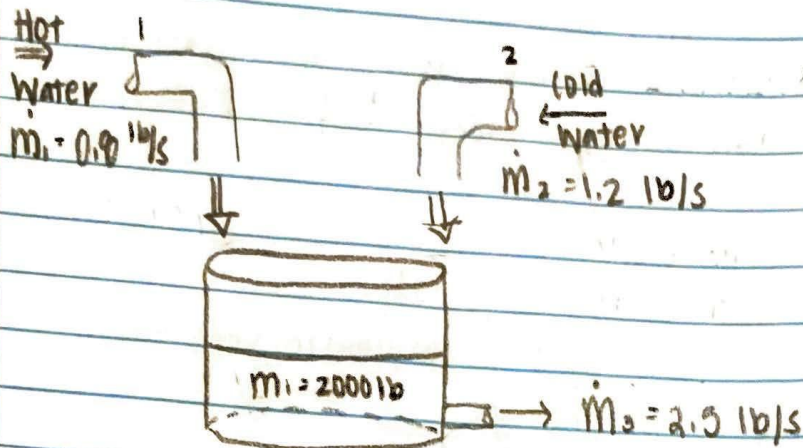
$$v = \frac{(0.314 \text{ m})^2 (274.92 \text{ K})}{(101325 \text{ Pa}) (20.97 \text{ kg})} = 0.770 \text{ m}^3/\text{kg}$$

... continues ...

2 Question 2 15 / 15

✓ - 0 pts Correct

#2.



$m_f = ??$ after one hour

$$\dot{m}_1 = (0.9 \text{ lb/s}) (60 \text{ s}) (60 \text{ min}) = 2900 \text{ lb/hr}$$

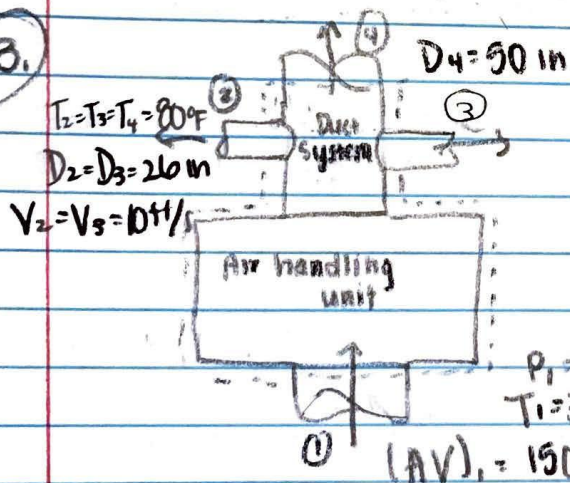
$$\dot{m}_2 = (1.2 \text{ lb/s}) (60 \text{ s}) (60 \text{ min}) = 4320 \text{ lb/hr}$$

$$\dot{m}_3 = (2.5 \text{ lb/s}) (60 \text{ s}) (60 \text{ min}) = 9000 \text{ lb/hr}$$

$$m_f = (2000 \text{ lb}) + (2900 \text{ lb/hr}) + (4320 \text{ lb/hr}) - (9000 \text{ lb/hr})$$

$$m_f = 200 \text{ lb} \leftarrow \text{amount of water after 1 hr}$$

#3.



Engineering model

① open system/control volume

② ideal gas

Part A: $\dot{m}_1 = \frac{AV}{v}$; $T = 35^\circ\text{F} \Rightarrow 274.02 \text{ K}$
 $P = 1 \text{ atm} \Rightarrow 101325 \text{ Pa}$

$$v = \frac{(0.314 \frac{\text{Nm}}{\text{K}})(274.02 \text{ K})}{(101325 \text{ Pa})(20.97 \text{ kg})} = 0.770 \text{ m}^3/\text{kg}$$

... continues ...

$$V = 0.778 \frac{\text{m}^3}{\text{kg}} \left| \frac{(3.28 \text{ ft})^3}{1 \text{ m}^3} \right| = 27.47 \frac{\text{ft}^3}{\text{kg}}$$

$$\dot{m}_1 = \frac{AV}{V} = \frac{(15,000 \text{ ft}^3/\text{min})}{(27.47 \text{ ft}^3/\text{kg})} = 546.11 \text{ kg}/\text{min}$$

$$546.11 \frac{\text{kg}}{\text{min}} \left| \frac{1 \text{ min}}{60 \text{ sec}} \right| \left| \frac{2.205 \text{ lb}}{1 \text{ kg}} \right| = 20.07 \frac{\text{lb}}{\text{s}}$$

Part B:

$V \cdot A \leftarrow$ volumetric flow rate

$$V = 10 \text{ ft}/\text{s} \quad r = 13 \text{ m} \left| \frac{1 \text{ ft}}{12 \text{ m}} \right| = \frac{13 \text{ ft}}{12}$$

$$\frac{10 \text{ ft}}{\text{s}} \left| \frac{60 \text{ s}}{1 \text{ min}} \right| = 600 \frac{\text{ft}}{\text{min}}$$

$$A = \pi r^2$$

$$A = \pi \left(\frac{13}{12} \right)^2 = 3.69 \text{ ft}^2$$

$$A \cdot V = 600 \frac{\text{ft}}{\text{min}} \cdot 3.69 \text{ ft}^2 = 2212.20 \frac{\text{ft}^3}{\text{min}}$$

Part C:

$$\dot{m}_1 - \dot{m}_2 - \dot{m}_3 - \dot{m}_4 = 0$$

$$A = \pi \left(\frac{25}{12} \right)^2 = 13.64 \text{ ft}^2$$

$$14.65 \frac{\text{lb}}{\text{s}} \left| \frac{1 \text{ kg}}{2.205 \text{ lb}} \right| = 6.64 \frac{\text{kg}}{\text{s}}$$

$$20.07 \frac{\text{lb}}{\text{s}} - 2.71 \frac{\text{lb}}{\text{s}} - 2.71 \frac{\text{lb}}{\text{s}} - \dot{m}_4 = 0 \quad \therefore \dot{m}_4 = 14.65 \frac{\text{lb}}{\text{s}}$$

$$v_2 = \frac{(9314)(299.002 \text{ K})}{(101325 \text{ Pa})(20.97 \text{ kg})} = 0.0049 \frac{\text{m}^3}{\text{kg}} \left| \frac{(3.28 \text{ ft})^3}{1 \text{ m}^3} \right| = 29.96 \quad \frac{2212.20}{29.96} =$$

$$6.64 \frac{\text{kg}}{\text{s}} = \frac{(13.64)(V)}{(29.96 \text{ ft}^3/\text{kg})}$$

$$V = 14.58 \frac{\text{ft}^3}{\text{s}}$$

3 Question 3 15 / 15

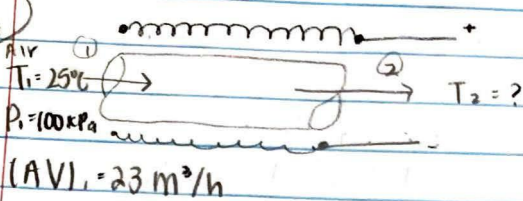
✓ - 0 pts Correct

- 2.5 pts a) Wrong calculation
- 2.5 pts a) Wrong equation
- 2.5 pts b) Wrong calculation
- 2.5 pts b) Wrong equation
- 2.5 pts c) Wrong calculation
- 2.5 pts c) Wrong equation

Engineering model

- ① open system/control volume
- ② $\Delta PE = \Delta KE = 0$

#4,



$$T_1 = 25^\circ\text{C} + 273.15 = 298.15\text{ K}$$

$$P_1 = 100\text{ kPa}$$

$$(AV)_1 = 23\text{ m}^3/\text{h}$$

$$V = 120\text{ V}$$

$$I = 4\text{ A}$$

$$C_p = 1.005\text{ kJ/kg}\cdot\text{K}$$

Part A:

$$\dot{m} = \frac{(AV)}{v}$$

$$v = \frac{(8314\text{ N}\cdot\text{m}) (298.15\text{ K})}{(1 \times 10^5\text{ Pa})}$$

$$(28.97\text{ kg})$$

$$\dot{m} = \frac{(23\text{ m}^3/\text{h})}{(0.856\text{ m}^3/\text{kg})}$$

$$v = 0.856\text{ m}^3/\text{kg}$$

$$26.98\text{ kg/h}$$

Part B:

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} [(h_1 - h_2) + \frac{(V_1^2 - V_2^2)}{2} + g(z_1 - z_2)]$$

$$\dot{W}_{cv} = \dot{m} (h_1 - h_2)$$

$$\dot{W}_{cv} = \dot{m} C_p (T_1 - T_2)$$

$$\dot{m} = \frac{26.98\text{ kg}}{\text{h}} \left| \frac{1\text{ hr}}{60\text{ min}} \right| \left| \frac{1\text{ min}}{60\text{ sec}} \right|$$

$$P = \dot{W}_{cv}$$

$$= 0.00746$$

$$P = \text{voltage} \cdot \text{current}$$

$$P = (120\text{ V})(4\text{ A}) = -480\text{ W} = -0.48\text{ kW}$$

$$-0.48\text{ kW} = (0.00746)(1.005\text{ kJ/kg}\cdot\text{K})(298.15 - T_2)$$

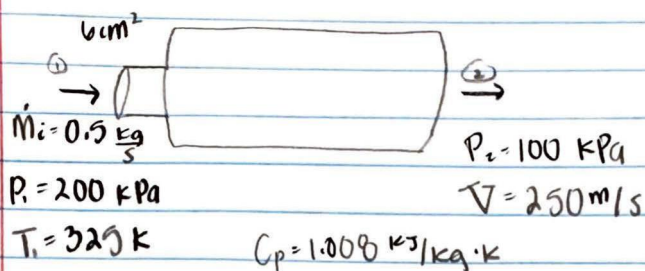
$$-T_2 = -362.12\text{ K}$$

$$T_2 = 90.97^\circ\text{C}$$

4 Question 4 20 / 20

✓ - 0 pts Correct

#5.



Engineering Model

- ① open system/control volume
- ② steady state
- ③ ideal gas
- ④ $\Delta PE = 0$

Part A:

$$v = \frac{(8314 \text{ N}\cdot\text{m}/\text{K})(325 \text{ K})}{(2 \times 10^5 \text{ Pa})} = 0.466 \frac{\text{m}^3}{\text{kg}}$$

$$\frac{(6 \text{ cm}^2)}{100 \text{ cm}} = 6 \times 10^{-4} \text{ m}$$

$$0.5 \text{ kg/s} = \frac{(6 \times 10^{-4} \text{ m})(V)}{(0.466 \text{ m}^3/\text{kg})}$$

$$V = 388.63 \text{ m/s}$$

Part B:

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \frac{(V_1^2 - V_2^2)}{2} + g(z_1 - z_2) \right]$$

$$h = C_p \cdot T$$

$$C_p(T_1 - T_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) = 0$$

$$(1.008 \text{ kJ/kg}\cdot\text{K})(325 \text{ K} - T_2) + \left(\frac{(388.63)^2 - (250)^2}{2} \right) = 0$$

$$1.008(325 - T_2) = \left[\frac{1}{1000} \right] \left(\frac{250^2 - 388.63^2}{2} \right)$$

$$T_2 = 368.92 \text{ K}$$

Part C:

$$\dot{m} = \rho_2 A_2 V_2$$

$$\dot{m}_1 = \dot{m}_2$$

$$\dot{m} = \frac{A_2 V_2}{V_2}$$

$$V_2 = \frac{(10314 \text{ N}\cdot\text{m}/\text{K})(340.92 \text{ K})}{(1 \times 10^5 \text{ Pa})(28.97 \text{ kg})} = 1.04 \text{ m}^3/\text{kg}$$

$$0.5 = \frac{A_2 (250 \text{ m/s})}{(1.04 \text{ m}^3/\text{kg})}$$

$$A_2 = 0.00211 \text{ m}^2 \Rightarrow 21.176 \text{ cm}^2$$

5 Question 5 20 / 20

✓ - 0 pts Correct

Roxanna Mendoza

CHE 201

March 10, 2020

#6. Oceans have various uses to them, some being a form of transportation, as a living source to aquatic living organisms, petroleum, and minerals. There is potential for oceans to be used to produce energy, power. There are two different types of technologies that are being researched regarding this development, one being tidal power technology and the other wave power technology.

Wave energy can be compared to the concept of solar power, instead wind is the source of energy. Air molecules, also known as wind, hit water molecules causing friction between the two forming waves. Wave energy is more concentrated than solar radiation which is then converted to wind energy, and finally to wave energy. Height and wave velocity are important factors when it comes to determining the energy of the waves. A way in order to harness waves they are channeled into a reservoir and then used to power turbines.

Tidal energy is based on the rise and fall of tides in the ocean. Tides are caused by the interaction between the ocean and earth's gravity pull. As well as waves the height and velocity of tides are important when it comes down to the amount of energy generated. One way in which they harness tides are that they are turbines in tidal streams which then takes energy from the flow of the water. Water is used in both methods because water is more dense than air therefore better results and more energy is gathered from the ocean.

Factors that have to be considered for both of these methods is that there are many living organisms located in the ocean and turbines are a potential harm to the organisms. Also depending on the size of the turbines they may be a harm to those ships that use the ocean as a source of transportation. Bulky machines usually disrupt the tides they are trying to gather which can be a negative factor. The construction of these machines can be costly as well. There are some weather conditions which can be damaging to the machines and the damages may be costly.

Some of the positive factors that can come with considering the ocean as a source of power would be that it is a type of renewable energy. This source of power is clean since there aren't any gases being released into the atmosphere which lead to pollution. When looking at the big picture they can be cost efficient once they are in place when compared to the price of the other sources of power such as coal or oil plants. The system is stored neatly under the ocean which is a plus.

References

National Geographic Society. (2012, October 9). tidal energy. Retrieved from

<https://www.nationalgeographic.org/encyclopedia/tidal-energy/>

Sarah. (2017, September 26). Harnessing the power of the ocean. Retrieved from

<https://www.science.org.au/curious/technology-future/ocean-power>

U.S. Energy Information Administration - EIA - Independent Statistics and Analysis. (n.d.).

Retrieved from <https://www.eia.gov/energyexplained/hydropower/wave-power.php>

6 Question 6 12.5 / 15

- 0 pts Correct

✓ - 2.5 pts Not enough evaluation on technical factors

- 5 pts Not enough evaluation on technical factors

- 2.5 pts Not enough evaluation on economic factors

- 5 pts Not enough evaluation on economic factors

- 2.5 pts Not enough evaluation on the effects on the ecosystem

- 5 pts Not enough evaluation on the effects on the ecosystem

- 15 pts Not answered